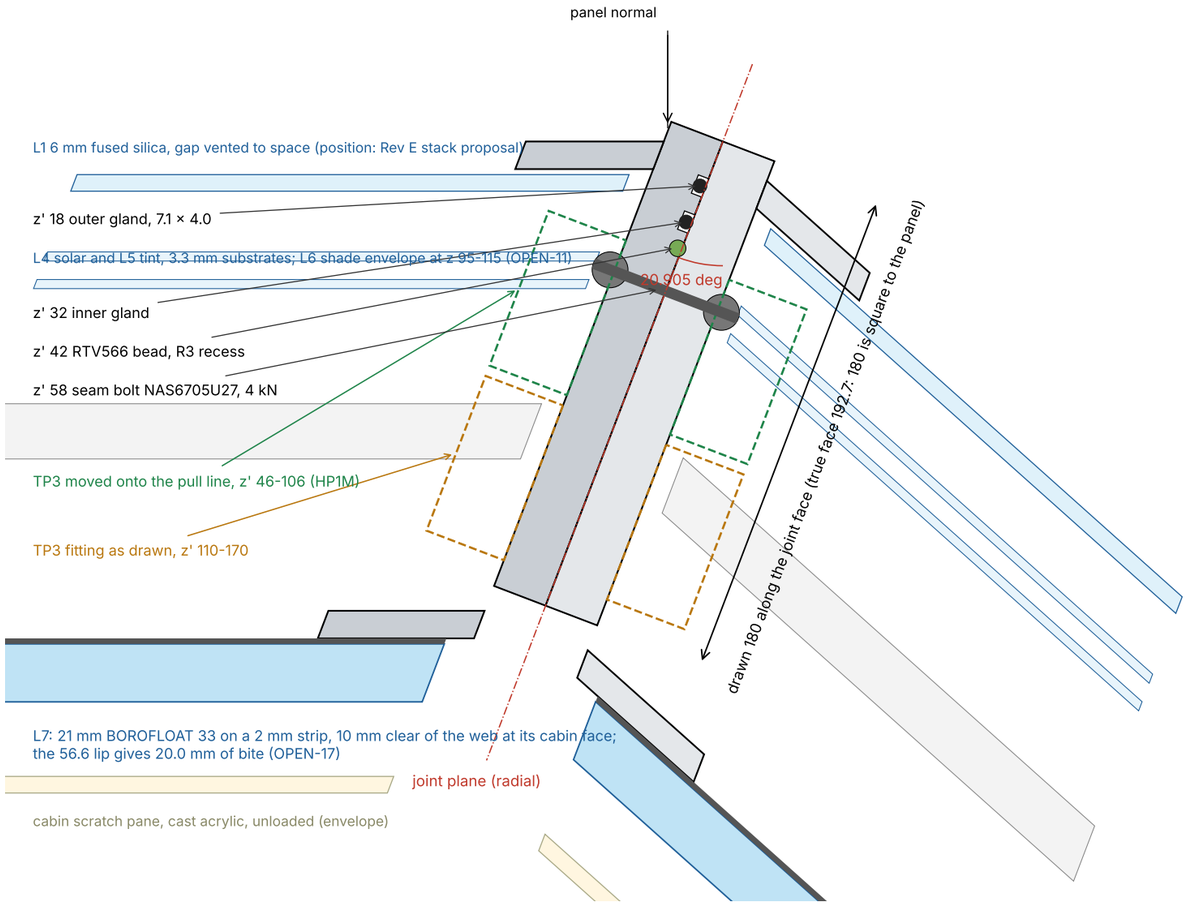
Section normal to a hexagon-to-hexagon seam at the first bolt (271.25 mm from the vertex). Space side at the top. Frame B (left) carries the glands.



Joint-face features, placed along the joint plane (z')

Panel depth = z' x cos(mitre): hexagon 0.9342, pentagon 0.9590. Both frames of a seam share the plane, so holes and glands line up on hexagon-to-pentagon seams too (OPEN-9).

Feature	z'	Hexagon z	Pentagon z
Outer gland	18	16.82	17.26
Leak port	25	23.35	23.97
Inner gland	32	29.89	30.69
RTV recess	42	39.24	40.28
Seam bolt row	58	54.18	55.62
TP3 pads, as drawn	110-170	102.8-158.8	105.5-163.0
TP3 pads, moved (HP1M)	46-106	43.0-99.0	44.1-101.6
Corner tie rows	133.8, 380.0	125.0, 355.0	128.3, 364.4

Seal: two loops of Parker BA500 butyl 5.33 mm in 7.1 × 4.0 glands (25 % squeeze, 79 % fill), RTV566 bead cured on the gro
Bolt: NAS6705U27 A286 with KFN670 nut, 4 kN by torque, 16 per seam at 75 pitch in bays of 3, 5, 5, 3; the two beside each
Loop corners: R16 minimum at the corner block, OPEN (Q92).

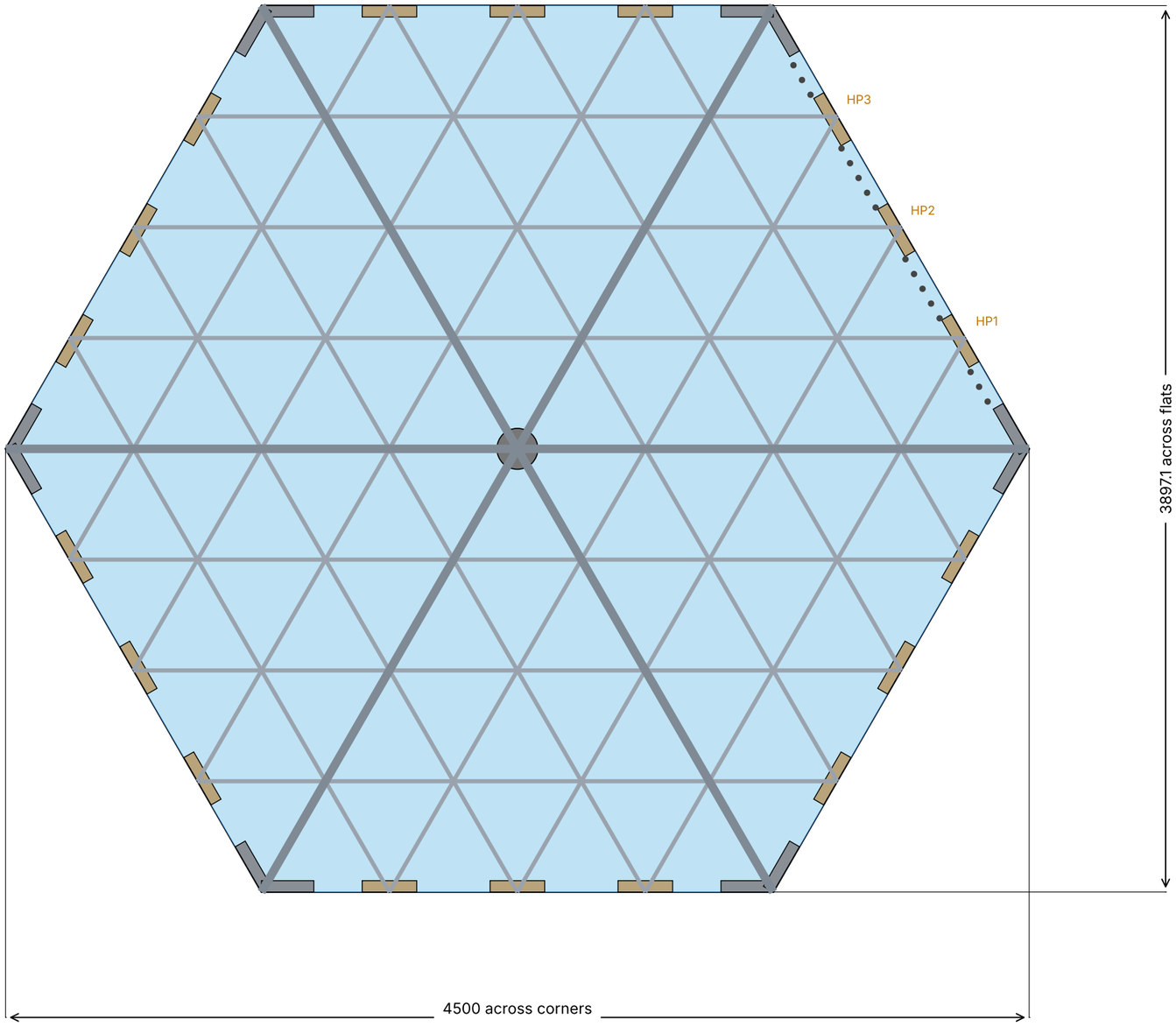
What changed from RE-04 (24 Oct): frame 140 to 180; pane 22 to 21 mm; bolt NAS6705U27 at 4 kN (was NAS9306C-06, a rivet standard); the lip and seat moved to the cabin face (z 170-180, P-06) for the restack (Q78);
the hardpoint shown in both positions because the fitting as drawn fails the HP1 seal check (0.22 atm free) and the moved fitting holds (0.064 mm at 1.93 atm); Rick agreed the move on 10 Nov. Shear along the seam goes to the hardpoint pins and the vertex dowels, no shear key (OPEN-15, confirmed by FEA 24 Oct). Secondary ribs (I 180 × 60 × 10) and the main rib cap angles (z 375-400) are off this section: see RF-01 and RF-06.

HCSA RF-05 Seam cross-section, hexagon to hexagon, at the first bolt

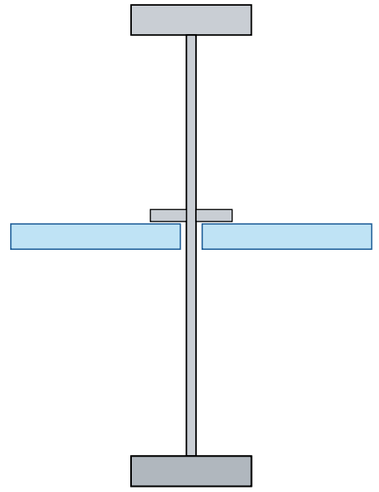
Rev F flight design, 9 Nov 2026. Replaces the Rev E draft sheet of 24 Oct	mm. Drawn from the parameters
Status: frame 180, gland, bolt decided 5 Nov; hardpoint position decided 10 Nov; outer layers OPEN-1	PROPOSAL where marked
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining

Hexagon from the cabin side (schematic, 1:20 on A3): frames, six main ribs, the k 4 secondary grid, hub, corner blocks (grey), hardpoint blocks (gold), 96 panes

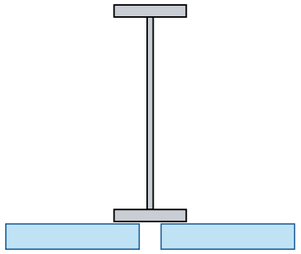
16 seam bolts per edge at 75 pitch in bays of 3, 5, 5, 3 (dots on the upper right edge); the first bolt 271.25 from the vertex; hardpoints HP at 562.5, 1125, 1687.5



Main rib B: I 400 × 100 × 25, web 8
seat ledges at z 170-180, cap angles P-04A at z 375-400



Secondary rib: I 180 × 60 × 10, web 5
pane seat on the cabin face z 180



21 mm panes on 2 mm setting strips, cut 9.09 mm short of the rib centre lines; bite 40.9 (main) and 20.9 (secondary); P-11 brace at every crossing (RF-08 r

k 4 grid: 562.5 mm pitch, 108 secondary rib segments, 96 panes (48 full triangles, 48 cut back at the frame and the hub), 18 six-way nodes, 18 crossings on the main ribs, 18 hardpoints

Parts list, one hexagon (model masses)

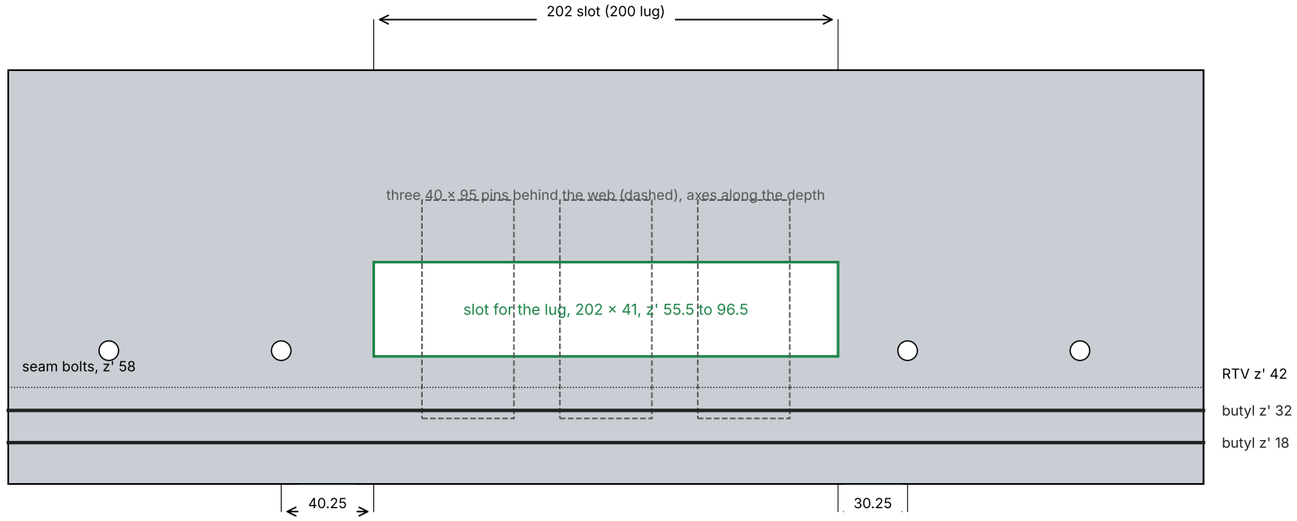
Part	Qty	Each kg	Total kg
P-06H frame member, 1,790 between the blocks	6	22.55	135
P-04 main rib B	6	33.24	199
P-04A cap angle	12	12.77	153
P-05 hub core	1	28.28	28
P-05A hub star plate	2	14.20	28
P-01 secondary rib segment	108	2.53	273
P-02 node star plate	36	1.75	63
P-03 node web core	18	4.38	79
P-07 pane, 21 mm	96	5.50	528
P-08 pane corner clamp	18	0.19	3
P-14 seam bolt (half of 16 × 6)	48	0.03	1
CB-01 corner block, leaning faces (9 Nov)	6	22.86	137
HP-02A lug body, Ti-6Al-4V (half of 18 hardpoints)	9	9.69	87
HP-02B clevis body, Ti-6Al-4V (the other half)	9	6.77	61
HP-03 pin 40 × 95, A286 (3 per hardpoint, shared)	27	0.95	26
P-11 main rib brace pair, 7075	18	1.12	20

Listed parts: 1824 kg per hexagon with its glass (528 kg) and its half share of the hardpoints. Masses at 6061 density; shell model 36.1

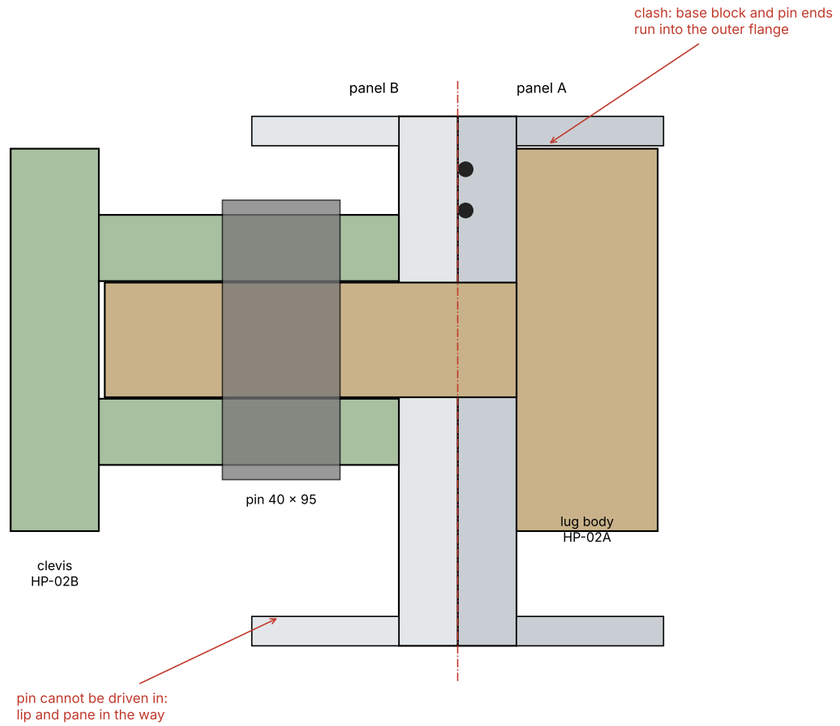
HCSA RF-06 Hexagon panel, general arrangement and parts list

Rev F flight design, 9 Nov 2026. Replaces the Rev E draft sheet of 24 Oct	mm. Drawn from the parameters
Status: ribs and frame decided; blocks, hardpoints, nodes PROPOSAL; model masses	PROPOSAL where marked
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining

Joint face at a hardpoint (HP1 of a seam): only the lug crosses the joint, through a slot on the cabin side of all three seal lines



Section across the seam, drawn straight (space side at the top): lug in two cheeks, pins in double shear



What the fitting is. Measured from the October C3 model (the run that gave 2.876 million N/mm): a Ti-6Al-4V lug 39 thick and 200 wide on panel A reaches across the joint into a clevis of two 22.5 mm cheeks on panel B; three A286 pins, 40 × 95, at 60 pitch, run along the panel depth with no clearance and work in double shear. The 8 November solids (pins across the joint, a pad of 200 × 60) were wrong and are withdrawn.

What the models say (HP1, 6-7 Nov). With the fitting where sheet RE-06-1 put it, 64 mm deeper than the line of the pull, the frames hinge about it and the seal gap reaches 0.11 mm at 0.22 atm (0.47 with the frames held) against the 1.93 atm required. With it on the line of the pull (lug mid-plane at z' 76) the gap at the hardpoint is 0.064 mm at 1.93 atm; the band at the plain bolt beside it reaches 0.11 mm at 1.87 atm at 4 kN, 3 % short. That bolt moved closer (115 mm) is worse, 1.76 atm (13 Nov); tightened to 6 kN it clears, about 1.95 atm (14 Nov), so the two bolts beside each hardpoint go to 6 kN, measured.
These runs model the fitting as a spring between rigid pads of 200 × 60, not this part.

What the solid found (9 Nov, leaning hexagon frame). Good: the slot is 10.5 mm from the RTV recess, 20 mm from the inner gland and 32 mm from the nearest bolt hole; the seam bolts on the flight layout (131.25 and 141.25 mm from the hardpoint centre) are clear of it; the 8 Nov 'pad covers the bolts' clash was a bolt drawn in the wrong place. Not good: (1) to sit behind the 20 mm webs the bodies had to move apart 38 mm, so the lug is 140 long, not 102, and the pin line is 60 mm from the joint, not 40: the C3 stiffness no longer applies. (2) The base blocks and pin ends run into the outer flange, and both bodies and the pins run into the two secondary ribs that land on the hardpoint. (3) The pins cannot be driven in: the seat lip and the pane are in the way from the cabin side, the flange from the space side. (4) The web is cut through over 202 mm at the bolt row's depth, and nothing yet says how the base blocks fasten to the frame. The old 50 mm local block (HP-01) has no job left. So the position is right and the part is not yet a design: it needs to be redrawn for the 180 mm frame (a shallower stack, pins put in from the side or along the seam, the ribs landing on the base blocks) and C3 and HP1 rerun on that. Load 108.7 kN per atm on the worst hardpoint; one hardpoint is 19.3 kg.

The three hardpoints on a 2,250 mm seam, from the vertex, with the nearest seam bolts (bays 3-5-5-3 at 75 pitch)

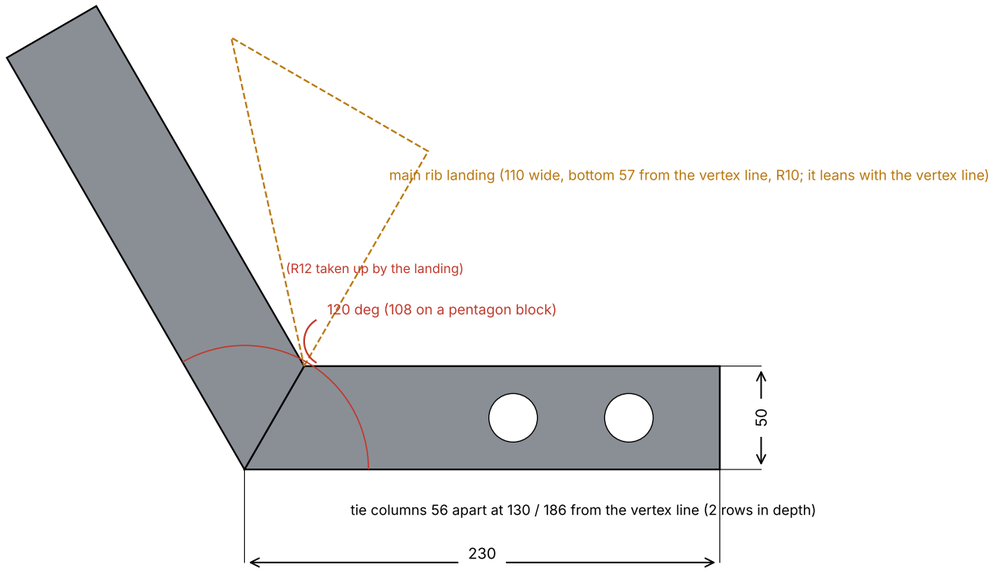
HP1 at 562.5: slot from 461.5 to 663.5; nearest bolts 421.25 (141.25 from the centre) and 693.75 (131.25 from the centre)

HP2 at 1125.0: slot from 1024.0 to 1226.0; nearest bolts 993.75 (131.25 from the centre) and 1256.25 (131.25 from the centre)

HP3 at 1687.5: slot from 1586.5 to 1788.5; nearest bolts 1556.25 (131.25 from the centre) and 1828.75 (141.25 from the centre)

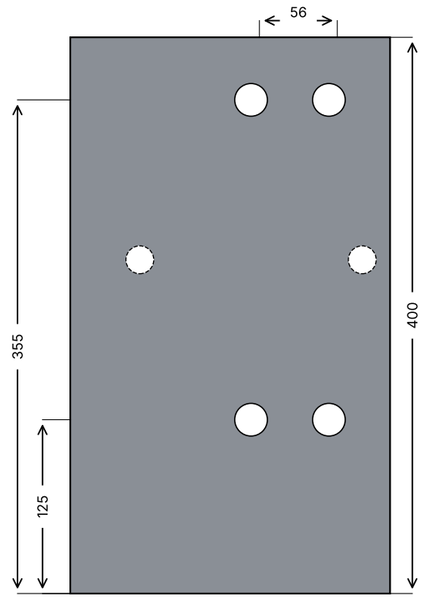
HCSA RF-07 Seam hardpoint: the TP3 fitting from the C3 model, at the moved position	
Rev F flight design, 9 Nov 2026. Replaces the Rev E draft sheet of 24 Oct	mm. Drawn from the parameters
Status: PROPOSAL, not yet buildable; position decided Tue 10 Nov	PROPOSAL where marked
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining

Plan of one hexagon corner block (CB-01), looking at the cabin face



joint faces: the legs' outer faces, on the radial joint planes

Joint face of a leg (z' 133.8 and 380 along the plane)



4 ties 7/8-14 Inconel 718 (23.5 holes), 2 dowels 20 (24 room)
hexagon rows z 125 / 355, pentagon rows z 128.3 / 364.4

What the corner model says (C1, 5 Nov)

- Tie preload 198 kN, held to 196.5-198.1 kN, by stretch
- Worst tie at proof, 0.8 atm: 214.5 kN (limit 227)
- Pentagon 15 K warmer: 213.3 kN
- Hatch open, 1 atm: 226.3 kN, 0.7 kN under the limit
- Faces open at proof: 0.55 mm (0.74 at 1 atm), limit 0.8
- Ties at 160 kN (14 Nov): 181.5 kN, faces 0.60 mm; 1 atm case to run
- Block stress, 95th percentile: 236 MPa (255 at 1 atm), no limit set yet
- Worst corner load 235 kN at 0.8 atm, 290 at 1 atm

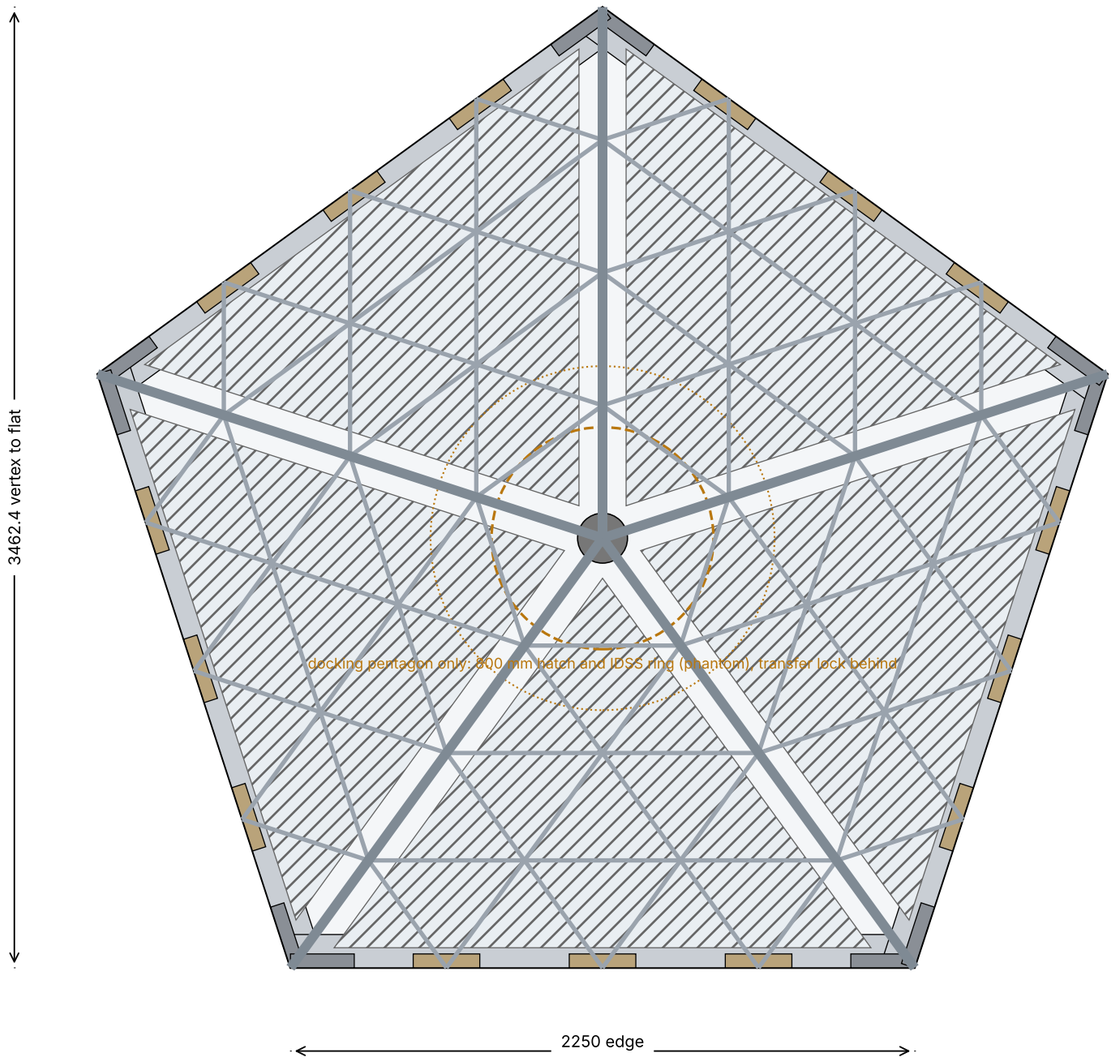
Making it (Marco, 7 Nov). As drawn the node cannot be machined: every inside corner needs a radius a cutter can reach (R10 here; the landing takes up the re-entrant corner), and the part could be cast instead of machined. Our position (9 Nov): machine it from 7050-T7451 or 2219-T851 plate, not a casting (seal faces) and not a die forging (180 parts). The main rib's flanges enter the block at the vertex landing (notch, PROPOSAL); the two frame members run into the legs' inner faces; the tie-to-web-fillet clash (Q29), the O10 maze-patch corner seal and the loop corners R16 (Q27, Q92) are open. The ties are tightened by measured stretch, because a torque wrench scatters about 25 % and the hatch-open case leaves only 0.7 kN. Forces Marco called 'achievable but unreasonable': 198 kN per tie (about 650-900 N m if torqued), 12 ties per node, 720 on the station. The corner node assembly A-03 (two hexagon blocks and one pentagon block at a real vertex, dihedral 138.19 / 142.62 deg) is in the parts library. Solids of 9 Nov, with the real leaning joint faces (CB-01 22.9 kg, CB-02 21.6 kg, node A-03 80.2 kg): three blocks placed at a real vertex share no volume, the joint faces are coplanar and the 12 tie holes and 6 dowel holes meet on axis. The block is one outline swept along the vertex line, which leans 23.8 deg (20.1 pentagon), so the main rib end is mitred to match. Tie columns 130 / 186 mm from the vertex line (87 / 143 cannot clear the landing). Metal from landing to joint face 20.4 mm (17.1 pentagon); washer seat to landing edge 14.3 mm (3.4 pentagon, tight). The R10 corners are 430 mm long: cut from the open side.

Main rib lateral brace (new requirement, 4 Nov): the main rib buckles sideways if unbraced over 2.25 m (0.79 of its load) and has 8.4 times margin braced every 562.5 mm. P-11 (PROPOSAL): a pair of 8 mm 7075 plates on the main rib web from z 180 to z 375 at every secondary rib crossing, bolted through the web, tying the cabin flange to the grid. Size to be checked.

HCSA RF-08 Corner node: blocks, ties, dowels, what the model says, machining

Rev F flight design, 9 Nov 2026. Replaces the Rev E draft sheet of 24 Oct	mm. Drawn from the parameters
Status: PROPOSAL; ties, preload decided; radii per Marco; O10 seal, rib landing open	PROPOSAL where marked
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining

Pentagon from the cabin side (schematic, 1:20 on A3): the same frame, main rib and secondary rib parts as the hexagon, no glass, a 10 mm closure plate



k 4 grid: 90 secondary rib segments, 5 main ribs, 5 corner blocks, 15 hardpoints; closure plate 10 mm 6061-T6 as five triangles (hatched), sealed at the main ribs

What the twelve pentagons do

Set by Manos on 25 Oct (PROPOSAL for TRUTH; the model and the earlier TRUTH split still show shield, service, docking and greenhouse pentagons):

- Eleven identical pentagons: inside, a greenhouse alcove and a service panel connected to the central mast; outside, solar panels on standoffs over an insulating blanket, and a stowed inflatable shield that cuts UV, takes heavy MMOD strikes and can act as a repair balloon (Q89).

- One docking pentagon: IDSS ring, 800 mm hatch, the transfer lock and relief valve (RF-04); the hatch stays shut whenever the pressures differ.

- Stabilising thrusters, if ever added, live in the pentagons (Q90). Which way the mast runs is open (Q88).

- The closure plate carries pressure only; a thin stressed skin would see 190 to 680 MPa. At 3.6 m across it is wider than plate comes, so it is five triangles sealed at the main ribs. Which side of the ribs, and how it is held, is OPEN with Marco (Q57). Drawn here on the cabin side at the seat plane, where the glass sits on a hexagon.

- Solar cells must stand off the structure: on bare pentagons in full sun the hexagon-pentagon difference could reach about 100 K (about 380 kN on a corner); on a blanket it is 2 to 15 K (1 Nov model).

- The greenhouses get no daylight: 6 to 9 kW of LED per TRUTH 4; the arrays and batteries for that are not sized.

Pentagon frame: P-06 at 180 deep with the 50 mm lip (pentagons carry no glass). Main ribs P-04 (five), hub P-05 with five arms (variant of th
Corner blocks CB-02 (108 deg, tie rows z 128.3 / 364.4). Hardpoints as RF-07. Mitre 16.472 deg; joint-face features by z' (RF-05 table).
The parts library carries the pentagon assembly A-04 and the closure triangle PL-01 (PROPOSAL, 8 Nov).

HCSA RF-09 Pentagon panel, general arrangement, roles, docking	
Rev F flight design, 9 Nov 2026. Replaces the Rev E draft sheet of 24 Oct	mm. Drawn from the parameters
Status: ribs and frame decided; closure plate side OPEN (Q57); roles PROPOSAL (25 Oct)	PROPOSAL where marked
Drawn: Claude for M. Theofanous Checked: - Released: -	Not for machining